

NILESH PATIL

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SUMMARY

AI Engineer and Python Backend with hands-on experience building scalable REST APIs using FastAPI and developing AI-powered applications with LLMs and Retrieval-Augmented Generation (RAG). Strong foundation in Data Structures Algorithms, backend development, machine learning, and deploying production-ready AI solutions.

EDUCATION

B.Tech in Artificial Intelligence and Machine Learning

2023 – 2027

R. C. Patel Institute of Technology, Shirpur, India

CGPA: 8.08

EXPERIENCE

AI Intern – Autom8x (A8X)

Jun 2026 – Jul 2026

San Jose, California, United States

- Developed AI-powered applications using Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and AI automation.
- Built FastAPI backend services and REST APIs for AI applications with API integration and data processing workflows.
- Implemented prompt engineering techniques and evaluated LLM responses for improved output quality.
- Worked with LangChain, Ollama, Hugging Face models, and vector databases for production-ready AI pipelines.

SKILLS

- **Languages:** Python, Java
- **Backend:** FastAPI, REST APIs, SQLAlchemy, Pydantic, JWT Authentication, CRUD APIs
- **AI/ML:** Scikit-learn, TensorFlow, Keras, Machine Learning, Deep Learning, NLP
- **Generative AI:** RAG, LangChain, LLM Integration, Prompt Engineering, Ollama, Hugging Face
- **Databases:** MySQL
- **Cloud & Tools:** Docker, AWS, Git, GitHub, CI/CD, DVC, Jupyter
- **Libraries:** NumPy, Pandas, Matplotlib
- **Core CS:** Data Structures & Algorithms, OOPS and DBMS

PROJECTS

SpaceAI RAG System – Astronomy QA Chatbot

2026

- Developed a Retrieval-Augmented Generation (RAG) chatbot capable of answering 100+ astronomy-related questions.
- Processed 15+ astronomy eBooks into 8,000+ vector embeddings using FAISS for semantic search.
- Integrated Llama 3.2 via Ollama with LangChain to generate context-aware responses.
- Built FastAPI REST APIs and Streamlit interface for seamless interaction.
- Optimized retrieval with Top-K similarity search to improve response relevance.
- **GitHub:** SpaceAI RAG System

Network Security ML Pipeline – Threat Detection System

2025

- Developed an end-to-end Machine Learning pipeline to classify network traffic as Normal or Threat.
- Automated data ingestion, preprocessing, feature engineering, model training, and evaluation.
- Achieved 94% classification accuracy using optimized ML models.
- Built FastAPI-based bulk prediction APIs for real-time CSV analysis.
- Containerized the application using Docker and deployed it on AWS EC2.
- Used AWS S3 and MongoDB for scalable data storage.
- **Live Demo:** Threat Detection System

ACHIEVEMENTS

- Solved **600+** Data Structures and Algorithms problems on LeetCode.
- Secured **2nd Place** in the DSA Olympiad at SVKM, Dhule.